THE SUMMANDS OF AN ALGEBRAIC SPACE PRESENTED AS A COPRODUCT OF SHEAVES ARE ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Base change keeps surjectivity and etaleness of a representable transformation. Lemma lemma-base-change-representable-transformations-property, used at proof lines 1262-1263. Makes U x_F F_i -> F_i surjective and etale.
- Base change of a coproduct along an open and closed summand is a scheme. Lemma lemma-coproduct-sheaves-open-and-closed, used at proof lines 1264-1265. Shows U x_F F_i is a scheme.
- Definition of an algebraic space. Definition definition-algebraic-space, used at proof lines 1266-1268. Is the condition verified at the end.

Source reference. Original definition 025Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2831–2831. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3808–3808. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0F15: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18785–18785. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $S \in \mathrm{Ob}(Sch_{fppf})$. Let F be an algebraic space over S. Given a set I and sheaves F_i on $\mathrm{Ob}((Sch/S)_{fppf})$, if $F \cong \coprod_{i \in I} F_i$ as sheaves, then each F_i is an algebraic space over S.*

Proof. The representability of $F \to F \times F$ implies that each diagonal morphism $F_i \to F_i \times F_i$ is representable (immediate from the definitions and the fact that $F \times_{(F \times F)} (F_i \times F_i) = F_i$). Choose a scheme U in $(Sch/S)_{fppf}$ and a surjective étale morphism $U \to F$ (this exist by hypothesis). The base change $U \times_F F_i \to F_i$ is surjective and étale by Lemma 02WL. On the other hand, $U \times_F F_i$ is a scheme

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by Lemma 0F15. Thus we have verified all the conditions in Definition 025Y and F_i is an algebraic space. □